

# DECO7230 – Prototype 2a Week 9 Test Plan

Tiny Worlds – Ladybird Anatomy Exploration

Target: ~5 min / participant

## 1. Test Purpose

Evaluate whether participants can independently navigate the insect selection flow, discover and interact with the three Ladybird anatomy targets, understand the multimodal learning feedback, recognise when the mission is complete, and provide actionable evidence for the design direction of Prototype 2b.

## 2. Research Questions and Assumptions

| ID       | Testable question                                                                                        | Assumption                                                                            |
|----------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| RQ1 / A1 | Can participants select the available insect and recognise unavailable “Coming Soon” insects?            | Selection flow is understandable without facilitator explanation.                     |
| RQ2 / A2 | Can participants discover antennae, elytra and flight wings without being told their names or locations? | Blue interaction cues and the mission framing are sufficient for discovery.           |
| RQ3 / A3 | Do glow, short cards, narration, animation and sound make interactions and functions understandable?     | Multimodal feedback supports both interaction clarity and immediate knowledge recall. |
| RQ4 / A4 | Is it clear when the Ladybird exploration mission is complete?                                           | Completion narration/feedback provides a clear endpoint.                              |
| RQ5 / A5 | What should the next prototype allow users to learn or physically do?                                    | Recurring preferences will identify a defensible Prototype 2b direction.              |

## 3. Validation Metrics and Success Criteria

| Aim                    | Measure                                     | Success criterion                                                                               |
|------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| A1 Selection           | Task success; Coming Soon recognition       | ≥4/5 select Ladybird and recognise an unavailable insect.                                       |
| A2 Discoverability     | 3 anatomy targets found; assistance level   | ≥4/5 find all 3. No participant should need more than one direct hint.                          |
| A3 Feedback / learning | 6 experience ratings; 3 knowledge questions | Relevant experience means ≥4/5 and ≥80% correct knowledge responses.                            |
| A4 Completion          | Completion reached; mission clarity rating  | ≥4/5 reach completion and mission clarity mean ≥4/5.                                            |
| A5 2b direction        | Preference counts + comments                | At least one recurring next-step theme appears in ≥2 participants and is supported by comments. |

## 4. Methodology and Rationale

- Method: short moderated usability test using a consistent task script. Observation captures actual behaviour; the knowledge check captures immediate understanding; 1–5 ratings capture perceived clarity/usability; open and multi-select questions generate evidence for Prototype 2b.
- Minimal intervention: participants are not told where the three anatomy targets are. This makes discoverability measurable rather than demonstrating the interaction for them.
- No forced think-aloud during exploration: narration and sound are part of the prototype, so continuous talking could mask the audio feedback being evaluated. Hesitation and spontaneous comments are recorded instead.
- Participants: target at least 5 in-class peers/tutor participants. Use participant IDs only. The convenience sample is appropriate for rapid iteration but is not treated as representative of the final 10–12-year-old target audience.

## 5. Standardised 5-Minute Procedure

| Time      | Facilitator action                                                                                                                                            | Record                        |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 0:00–0:30 | Fit headset / confirm controller is working. Say: “Explore the prototype naturally. I won’t guide you unless you get stuck.”                                  | Start time; setup issue       |
| 0:30–3:00 | Participant uses Selection and explores Ladybird. Do not name the three body parts. Observe interaction sequence, hesitation, failed attempts and assistance. | Task ticks; hints; hesitation |
| 3:00–4:00 | Remove/relax headset if needed. Ask the 3 knowledge questions and 6 experience ratings exactly as written.                                                    | Responses; 1–5 ratings        |
| 4:00–5:00 | Ask the Prototype 2b questions and invite one final comment.                                                                                                  | Preference counts; quote      |

## 6. Assistance / Hint Rule

- Level 0 – No help: participant continues independently.
- Level 1 – Small hint after ~15 seconds of being stuck: “Look around the Ladybird for an interactive cue / blue glow.” Do not name or point to the body part.
- Level 2 – Direct help only if necessary to continue the session. Record as 2+ hints and note exactly what was explained.
- Stop or pause immediately if the participant reports discomfort, nausea, disorientation or wants to remove the headset.

## 7. Data Collection and Analysis Plan

| Data source           | Recorded as                                   | How it will be used                                                                                       |
|-----------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Task performance      | Yes/No completion + hint count                | Calculate success rate for selection, each anatomy target and completion; count assistance levels.        |
| Knowledge             | 3 short-answer questions                      | Score correct/incorrect; report total correct responses and recurring misconceptions.                     |
| Experience            | 6 Likert items (1–5)                          | Calculate mean for each item; flag ratings ≤3 for qualitative follow-up.                                  |
| Observation           | Hesitation, confusion, useful feedback, quote | Group repeated observations into themes (discoverability, controls, comfort, feedback).                   |
| Prototype 2b evidence | 3 next-step questions                         | Count each selected option; combine with open comments to justify the next interaction/content direction. |
| XR principles         | 5 principle ratings + issue notes             | Report mean and any safety/usability issue even when the average is high.                                 |

## 8. Facilitator Controls for Consistency

- Use the same opening script, hint timing and question wording for every participant.
- Do not praise a specific answer during the task or explain the correct anatomy before the knowledge check.
- Record behaviour first; interpretation happens after all sessions to reduce confirmation bias.
- If a technical failure occurs, note it separately from a usability failure.

## 9. Five XR Design Principles Check (after all sessions)

| Principle          | What to evaluate                                                    | 1–5       | Evidence / issue |
|--------------------|---------------------------------------------------------------------|-----------|------------------|
| Text Readability   | Can cards and selection text be read quickly at viewing distance?   | 1 2 3 4 5 |                  |
| Ergonomics         | Are pointing, viewing height, and interaction duration comfortable? | 1 2 3 4 5 |                  |
| Brightness Changes | Is blue glow noticeable without being harsh?                        | 1 2 3 4 5 |                  |
| Simulator Sickness | Any discomfort, disorientation, or visual instability?              | 1 2 3 4 5 |                  |
| Button Placement   | Are Start / selection / Back controls easy to locate and activate?  | 1 2 3 4 5 |                  |

## 10. Decision Rule for Prototype 2b

Prototype 2b should not simply repeat the anatomy point-and-select loop. The next direction will be chosen from recurring user needs (e.g., feeding, habitat or behaviour/movement) and must be supported by both preference counts and observed/quoted evidence. Existing feedback mechanisms will only be retained if the Week 9 data shows they support clarity without creating distraction or discomfort.

# Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: \_\_1\_\_

Researcher records A and D. Participant answers B and C after the task.

## A. Task Performance — Researcher observation

| Measure                            | Record                          |
|------------------------------------|---------------------------------|
| Selected Ladybird successfully?    | <input type="checkbox"/> Yes    |
| Help needed?                       | <input type="checkbox"/> 1 hint |
| Understood a 'Coming Soon' insect? | <input type="checkbox"/> Yes    |
| Antennae discovered?               | <input type="checkbox"/> Yes    |
| Elytra discovered?                 | <input type="checkbox"/> Yes    |
| Flight wings discovered?           | <input type="checkbox"/> Yes    |
| Reached completion message?        | <input type="checkbox"/> Yes    |

## B. Quick Knowledge Check — Short answers

| Question                                   | Participant response                 | Correct?                     |
|--------------------------------------------|--------------------------------------|------------------------------|
| What do the antennae help the ladybird do? | chemoreception                       | <input type="checkbox"/> Yes |
| What is the job of the elytra?             | Protect the more delicate hind wings | <input type="checkbox"/> Yes |
| What are the flight wings used for?        | Fly                                  | <input type="checkbox"/> Yes |

## C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

| Statement                                                             | 1                     | 2                     | 3                     | 4                     | 5                     |
|-----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| I understood what I was expected to do.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The blue glow made interactive parts easy to find.                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The card + voice explanation was easy to understand.                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The animations and sounds made my actions feel clear.                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| Overall, the prototype was easy to use.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The Ladybird exploration mission was easy to understand and complete. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |

## D. Observation & Prototype 2b Evidence

|                                                            |                                                                                                                                                                                                    |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Where did the participant hesitate?                        | Change perspective                                                                                                                                                                                 |
| What caused confusion?                                     | 3D vertigo                                                                                                                                                                                         |
| Which feedback helped most?                                | <input type="checkbox"/> Animation                                                                                                                                                                 |
| What should you be able to do next?                        | <input type="checkbox"/> Feeding <input type="checkbox"/> Habitat                                                                                                                                  |
| What would you most like to learn about the ladybird next? | <input type="checkbox"/> What it eats <input type="checkbox"/> Where it lives                                                                                                                      |
| What would make the next prototype feel more interactive?  | <input type="checkbox"/> Feeding the ladybird <input type="checkbox"/> Choosing/changing its habitat <input type="checkbox"/> Guiding its movement <input type="checkbox"/> Environmental response |

Participant quote / other comment: “\_\_I didn’t notice what the background voice is talking about\_\_”

| Principle          | What to evaluate                                                              | 1–5 | Evidence / issue |
|--------------------|-------------------------------------------------------------------------------|-----|------------------|
| Text Readability   | Can cards and selection text be read quickly at viewing distance?             | 4   |                  |
| Ergonomics         | Are pointing, viewing height, and interaction duration comfortable?           | 5   |                  |
| Brightness Changes | Is blue glow noticeable without being harsh?                                  | 5   |                  |
| Simulator Sickness | Any discomfort, disorientation, or visual instability? Note simulator limits. | 5   |                  |
| Button Placement   | Are Start / selection / Back controls easy to locate and activate?            | 5   |                  |

# Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: \_\_2\_\_

Researcher records A and D. Participant answers B and C after the task.

## A. Task Performance — Researcher observation

| Measure                            | Record                          |
|------------------------------------|---------------------------------|
| Selected Ladybird successfully?    | <input type="checkbox"/> Yes    |
| Help needed?                       | <input type="checkbox"/> 1 hint |
| Understood a 'Coming Soon' insect? | <input type="checkbox"/> Yes    |
| Antennae discovered?               | <input type="checkbox"/> Yes    |
| Elytra discovered?                 | <input type="checkbox"/> Yes    |
| Flight wings discovered?           | <input type="checkbox"/> Yes    |
| Reached completion message?        | <input type="checkbox"/> Yes    |

## B. Quick Knowledge Check — Short answers

| Question                                   | Participant response     | Correct?                     |
|--------------------------------------------|--------------------------|------------------------------|
| What do the antennae help the ladybird do? | Touch and smell          | <input type="checkbox"/> Yes |
| What is the job of the elytra?             | Protect the flight wings | <input type="checkbox"/> Yes |
| What are the flight wings used for?        | Fly                      | <input type="checkbox"/> Yes |

## C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

| Statement                                                             | 1                     | 2                     | 3                     | 4                     | 5                     |
|-----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| I understood what I was expected to do.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The blue glow made interactive parts easy to find.                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The card + voice explanation was easy to understand.                  | <input type="radio"/> | <input type="radio"/> | 3                     | <input type="radio"/> | <input type="radio"/> |
| The animations and sounds made my actions feel clear.                 | <input type="radio"/> | <input type="radio"/> | 3                     | <input type="radio"/> | <input type="radio"/> |
| Overall, the prototype was easy to use.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The Ladybird exploration mission was easy to understand and complete. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |

## D. Observation & Prototype 2b Evidence

|                                                            |                                                                                             |
|------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Where did the participant hesitate?                        | Didn't notice the voice guidance.                                                           |
| What caused confusion?                                     | Don't know what to do next                                                                  |
| Which feedback helped most?                                | <input type="checkbox"/> Animation                                                          |
| What should you be able to do next?                        | <input type="checkbox"/> Movement/behaviour                                                 |
| What would you most like to learn about the ladybird next? | <input type="checkbox"/> What it eats <input type="checkbox"/> How it behaves/              |
| What would make the next prototype feel more interactive?  | <input type="checkbox"/> Feeding the ladybird <input type="checkbox"/> Guiding its movement |

Participant quote / other comment: “\_\_ I feel sick when the ladybird come so close to me \_\_”

| Principle          | What to evaluate                                                              | 1–5 | Evidence / issue     |
|--------------------|-------------------------------------------------------------------------------|-----|----------------------|
| Text Readability   | Can cards and selection text be read quickly at viewing distance?             | 4   |                      |
| Ergonomics         | Are pointing, viewing height, and interaction duration comfortable?           | 4   |                      |
| Brightness Changes | Is blue glow noticeable without being harsh?                                  | 4   |                      |
| Simulator Sickness | Any discomfort, disorientation, or visual instability? Note simulator limits. | 3   | Too close to the bug |

|                  |                                                                    |   |  |
|------------------|--------------------------------------------------------------------|---|--|
| Button Placement | Are Start / selection / Back controls easy to locate and activate? | 4 |  |
|------------------|--------------------------------------------------------------------|---|--|

# Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: \_\_3 s\_\_

Researcher records A and D. Participant answers B and C after the task.

## A. Task Performance — Researcher observation

| Measure                            | Record                          |
|------------------------------------|---------------------------------|
| Selected Ladybird successfully?    | <input type="checkbox"/> Yes    |
| Help needed?                       | <input type="checkbox"/> 1 hint |
| Understood a 'Coming Soon' insect? | <input type="checkbox"/> Yes    |
| Antennae discovered?               | <input type="checkbox"/> Yes    |
| Elytra discovered?                 | <input type="checkbox"/> Yes    |
| Flight wings discovered?           | <input type="checkbox"/> Yes    |
| Reached completion message?        | <input type="checkbox"/> Yes    |

## B. Quick Knowledge Check — Short answers

| Question                                   | Participant response | Correct?                     |
|--------------------------------------------|----------------------|------------------------------|
| What do the antennae help the ladybird do? | Touch and smell      | <input type="checkbox"/> Yes |
| What is the job of the elytra?             | Protect the wings    | <input type="checkbox"/> Yes |
| What are the flight wings used for?        | Use to fly           | <input type="checkbox"/> Yes |

## C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

| Statement                                                             | 1                     | 2                     | 3                     | 4                     | 5                     |
|-----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| I understood what I was expected to do.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The blue glow made interactive parts easy to find.                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The card + voice explanation was easy to understand.                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The animations and sounds made my actions feel clear.                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| Overall, the prototype was easy to use.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The Ladybird exploration mission was easy to understand and complete. | <input type="radio"/> | <input type="radio"/> | 3                     | <input type="radio"/> | <input type="radio"/> |

## D. Observation & Prototype 2b Evidence

|                                                            |                                                                                                                  |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Where did the participant hesitate?                        | It's hard to find the wings can interact with                                                                    |
| What caused confusion?                                     | No                                                                                                               |
| Which feedback helped most?                                | <input type="checkbox"/> Glow <input type="checkbox"/> Animation                                                 |
| What should you be able to do next?                        | <input type="checkbox"/> Feeding <input type="checkbox"/> Other: __ Control movement like make the ladybird fly. |
| What would you most like to learn about the ladybird next? | <input type="checkbox"/> Where it lives                                                                          |
| What would make the next prototype feel more interactive?  | <input type="checkbox"/> Feeding the ladybird <input type="checkbox"/> Guiding its movement                      |

Participant quote / other comment: “\_\_ It would be great if can feed the insect \_\_”

| Principle          | What to evaluate                                                    | 1–5 | Evidence / issue |
|--------------------|---------------------------------------------------------------------|-----|------------------|
| Text Readability   | Can cards and selection text be read quickly at viewing distance?   | 4   |                  |
| Ergonomics         | Are pointing, viewing height, and interaction duration comfortable? | 5   |                  |
| Brightness Changes | Is blue glow noticeable without being harsh?                        | 5   | yes              |

|                    |                                                                               |   |          |
|--------------------|-------------------------------------------------------------------------------|---|----------|
| Simulator Sickness | Any discomfort, disorientation, or visual instability? Note simulator limits. | 5 | Not sick |
| Button Placement   | Are Start / selection / Back controls easy to locate and activate?            | 5 |          |

# Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: \_\_4\_\_

Researcher records A and D. Participant answers B and C after the task.

## A. Task Performance — Researcher observation

| Measure                            | Record                        |
|------------------------------------|-------------------------------|
| Selected Ladybird successfully?    | <input type="checkbox"/> Yes  |
| Help needed?                       | <input type="checkbox"/> None |
| Understood a 'Coming Soon' insect? | <input type="checkbox"/> Yes  |
| Antennae discovered?               | <input type="checkbox"/> Yes  |
| Elytra discovered?                 | <input type="checkbox"/> Yes  |
| Flight wings discovered?           | <input type="checkbox"/> Yes  |
| Reached completion message?        | <input type="checkbox"/> Yes  |

## B. Quick Knowledge Check — Short answers

| Question                                   | Participant response | Correct?                     |
|--------------------------------------------|----------------------|------------------------------|
| What do the antennae help the ladybird do? | sense the smell      | <input type="checkbox"/> Yes |
| What is the job of the elytra?             | Protect the wings    | <input type="checkbox"/> Yes |
| What are the flight wings used for?        | Fly                  | <input type="checkbox"/> Yes |

## C. Participant Experience — Circle one number

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

| Statement                                                             | 1                     | 2                     | 3                     | 4                     | 5                     |
|-----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| I understood what I was expected to do.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The blue glow made interactive parts easy to find.                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The card + voice explanation was easy to understand.                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The animations and sounds made my actions feel clear.                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| Overall, the prototype was easy to use.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |
| The Ladybird exploration mission was easy to understand and complete. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |

## D. Observation & Prototype 2b Evidence

|                                                            |                                                                                                      |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Where did the participant hesitate?                        | No hesitate                                                                                          |
| What caused confusion?                                     | The function part of the trigger need to change                                                      |
| Which feedback helped most?                                | <input type="checkbox"/> Glow <input type="checkbox"/> Voice <input type="checkbox"/> Sound          |
| What should you be able to do next?                        | <input type="checkbox"/> Feeding <input type="checkbox"/> Habitat                                    |
| What would you most like to learn about the ladybird next? | <input type="checkbox"/> What it eats <input type="checkbox"/> Where it lives                        |
| What would make the next prototype feel more interactive?  | <input type="checkbox"/> Feeding the ladybird <input type="checkbox"/> Choosing/changing its habitat |

Participant quote / other comment: “\_\_It could be very interesting if I can feed the lady bird or ride the ladybird\_\_”

| Principle        | What to evaluate                                                    | 1–5 | Evidence / issue |
|------------------|---------------------------------------------------------------------|-----|------------------|
| Text Readability | Can cards and selection text be read quickly at viewing distance?   | 5   |                  |
| Ergonomics       | Are pointing, viewing height, and interaction duration comfortable? | 5   |                  |

|                    |                                                                               |   |  |
|--------------------|-------------------------------------------------------------------------------|---|--|
| Brightness Changes | Is blue glow noticeable without being harsh?                                  | 5 |  |
| Simulator Sickness | Any discomfort, disorientation, or visual instability? Note simulator limits. | 4 |  |
| Button Placement   | Are Start / selection / Back controls easy to locate and activate?            | 4 |  |

# Prototype 2a – Usability Test & Observation Sheet

Tiny Worlds – Ladybird Exploration

Participant ID: \_\_5\_\_

Researcher records A and D. Participant answers B and C after the task.

## A. Task Performance — Researcher observation

| Measure                            | Record                        |
|------------------------------------|-------------------------------|
| Selected Ladybird successfully?    | <input type="checkbox"/> Yes  |
| Help needed?                       | <input type="checkbox"/> None |
| Understood a 'Coming Soon' insect? | <input type="checkbox"/> Yes  |
| Antennae discovered?               | <input type="checkbox"/> Yes  |
| Elytra discovered?                 | <input type="checkbox"/> Yes  |
| Flight wings discovered?           | <input type="checkbox"/> Yes  |
| Reached completion message?        | <input type="checkbox"/> Yes  |

## B. Quick Knowledge Check — Short answers

| Question                                   | Participant response  | Correct?                     |
|--------------------------------------------|-----------------------|------------------------------|
| What do the antennae help the ladybird do? | Sense touch and smell | <input type="checkbox"/> Yes |
| What is the job of the elytra?             | Protect wings         | <input type="checkbox"/> Yes |
| What are the flight wings used for?        | Fly                   | <input type="checkbox"/> Yes |

## C. Participant Experience — Circle one number

1 = Strongly disagree   2 = Disagree   3 = Neutral   4 = Agree   5 = Strongly agree

| Statement                                                             | 1                     | 2                     | 3                     | 4                     | 5                     |
|-----------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| I understood what I was expected to do.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The blue glow made interactive parts easy to find.                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The card + voice explanation was easy to understand.                  | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The animations and sounds made my actions feel clear.                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| Overall, the prototype was easy to use.                               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 5                     |
| The Ladybird exploration mission was easy to understand and complete. | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | 4                     | <input type="radio"/> |

## D. Observation & Prototype 2b Evidence

|                                                            |                                                                                                                                                    |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Where did the participant hesitate?                        | Did find any part hesitate.                                                                                                                        |
| What caused confusion?                                     | The design of the buttons on the trigger is a bit confuse.                                                                                         |
| Which feedback helped most?                                | <input type="checkbox"/> Glow <input type="checkbox"/> Animation <input type="checkbox"/> Voice                                                    |
| What should you be able to do next?                        | <input type="checkbox"/> Feeding <input type="checkbox"/> Habitat                                                                                  |
| What would you most like to learn about the ladybird next? | <input type="checkbox"/> What it eats <input type="checkbox"/> Where it lives                                                                      |
| What would make the next prototype feel more interactive?  | <input type="checkbox"/> Feeding the ladybird <input type="checkbox"/> Choosing/changing its habitat <input type="checkbox"/> Guiding its movement |

| Principle        | What to evaluate                                                  | 1–5 | Evidence / issue |
|------------------|-------------------------------------------------------------------|-----|------------------|
| Text Readability | Can cards and selection text be read quickly at viewing distance? | 5   |                  |

|                    |                                                                               |   |                            |
|--------------------|-------------------------------------------------------------------------------|---|----------------------------|
| Ergonomics         | Are pointing, viewing height, and interaction duration comfortable?           | 5 |                            |
| Brightness Changes | Is blue glow noticeable without being harsh?                                  | 4 | The wing is hard to notice |
| Simulator Sickness | Any discomfort, disorientation, or visual instability? Note simulator limits. | 5 |                            |
| Button Placement   | Are Start / selection / Back controls easy to locate and activate?            | 5 |                            |